

FULL NAME: _____

BLOCK: _____

Real + Complex Zeros, and Modeling

Lesson 3-5 · Period 3 · Student Packet

OBJECTIVES

Math Objective	Find all zeros (real and complex) of a polynomial function, and use the factored form to model a volume-constrained application.
Language Objective	Students will use the frame “The zeros are ___; because ___, the complex conjugate ___ must also be a zero.” to justify complex-zero reasoning.
Essential Understanding	Complex zeros of a polynomial with real coefficients come in conjugate pairs. A cubic model of a physical quantity (volume, profit) can be solved for specific values by factoring and solving.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students find all zeros (real + complex) of two cubics, then model a Storage Box volume problem end-to-end. Every item is Savvas-sourced.
Essential Question	How do real and complex zeros together describe a polynomial, and how do you use that structure to answer a real-world model?
Materials	Pencils; student packet; calculator (optional). Desmos allowed for Try-It 3 check only, NOT for Practice #27.

LAUNCH — Complex Zeros + Modeling (teacher models on screen)**[BOOK] COMPLEX-ZERO RULES (keep printed on your page)**

- A polynomial with real coefficients that has complex zeros has them in CONJUGATE PAIRS.
- If $a + bi$ is a zero, then $a - bi$ must also be a zero.
- A degree- n polynomial has exactly n zeros (counting multiplicity). Some may be real, some complex.
- Example: if a cubic has a real zero at $x = 4$ and a complex zero $2 + 3i$, its third zero must be $2 - 3i$.

[RULER] MODELING RULES (for the Storage Box)

- Set up a factored-form volume function $V(x) = (\text{expr}_1)(\text{expr}_2)(\text{expr}_3)$.
- Each factor is a physical dimension — label each: length, width, height.
- Use the physical constraint (e.g., “length is greatest”) to decide which factor is which dimension.
- To find dimensions at $V = 10 \text{ ft}^3$, solve the cubic equation — show all work.

EXPLORE — Think-Pair-Share (38 min)

Work with a partner. Use the rule cards above. Storage Box (Practice #27) is the big one — plan 20 min for it.

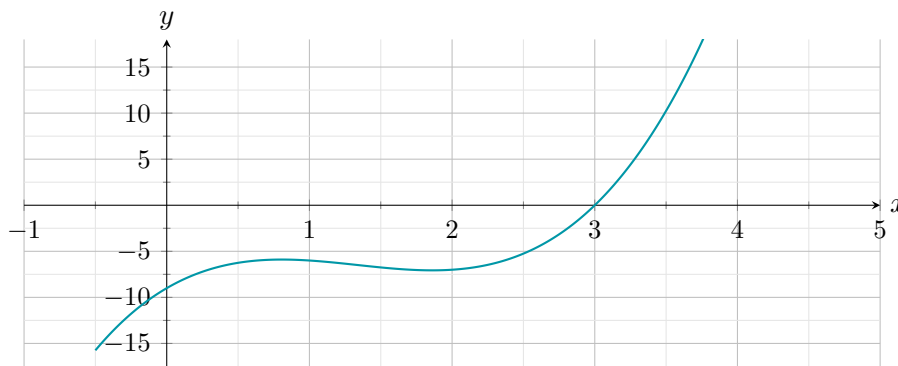
Bank-item labels (in order):

1. Try It 3a

2. Try It 3b
3. Practice #17
4. Practice #27 [STAR] MODELING

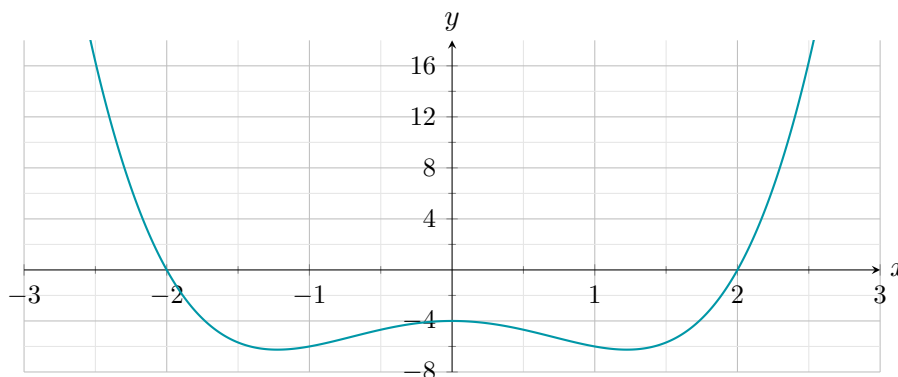
Try It 3a

What are all the real and complex zeros of the polynomial function shown in the graph? $f(x) = 2x^3 - 8x^2 + 9x - 9$.



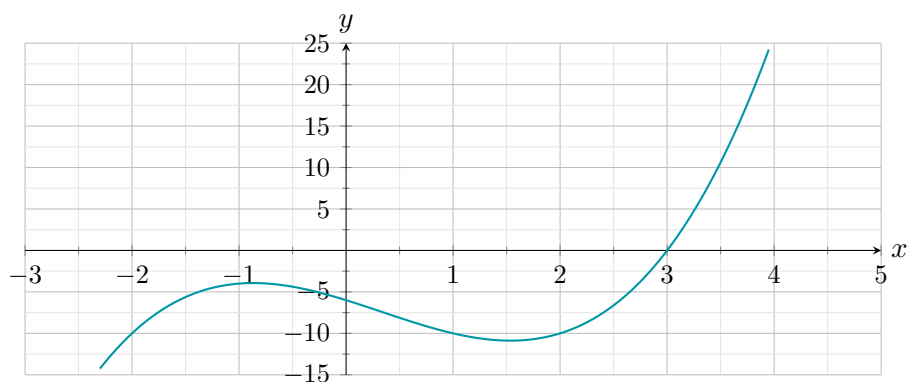
Try It 3b

What are all the real and complex zeros of the polynomial function shown in the graph? $f(x) = x^4 - 3x^2 - 4$.



Practice #17

What are all the real and complex zeros of the polynomial function shown in the graph? $f(x) = x^3 - x^2 - 4x - 6$.



Practice #27 [STAR] MODELING

Model With Mathematics: The height of a rectangular storage box is less than both its length and width. The function $f(x) = x^3 + 2x^2 - 3x$ represents the volume of the rectangular box, where x represents the width of the box, in feet.

- Find the factored form of $f(x)$.
- Find the zeros of the function.
- You know x represents the width of the box. What do the other two factors represent?
- Find the dimensions of the box when the volume is 10 ft^3 .

SHARE / SUMMARY (7 min)

Sentence frames

Pair share: “For the Storage Box, the factored form was _____. I chose _____ as the length because _____.”
To find $V = 10 \text{ ft}^3$, I _____.”

Whole class: one pair at random presents Part (a)-(b)-(c)-(d) of #27.

EXIT

Today I learned that
For the Storage Box, the hardest step was
Thursday’s assessment will test

OPTIONAL REINFORCEMENT (not collected)

If you finish early or want extra practice before Thursday's assessment.

Optional #1 (Savvas Practice)

Complete each statement so it means the same as “4 is a zero of the function”: (1) The graph of the function crosses the _____ at 4. (2) _____ is a factor of the polynomial.

Optional #2 (Savvas Practice)

Performance Task: Venetta opened several deli sandwich franchises in 2000. The profit P (in hundreds of dollars) of the franchises in t years (since the franchises opened) can be modeled by the function $P(t) = t^3 + t^2 - 6t$.

Part A: Sketch a graph of the function.

Part B: Based on the model, during what years did Venetta not make a profit?

Part C: If the model is appropriate, predict the amount of profit Venetta will receive from her franchises in 2020.